

Moein Razavi

AI/ML Engineer | Generative AI, Forecasting & Cloud Applications

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Professional Summary

AI/ML engineer and data scientist building AI applications and energy decision systems at Ford. Connects machine learning with deployed software: document RAG, text-to-SQL, utility-data ingestion, and forecasting for electricity-market bids. Brings Python/PyTorch expertise, cloud deployment experience, and full-stack engineering across energy and wearable health technology.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **EV analytics assistant:** Reduced time spent answering recurring analytics and document questions by 30% through natural-language access to BigQuery, legal documents, and vehicle manuals. Designed LangChain/GPT-4o orchestration for BM25-based retrieval-augmented generation (RAG) and schema-aware text-to-SQL agents, with intent routing, conversational context, and domain-rule answer checks; deployed Dash on Cloud Run with SAML/ADFS authentication.
- **Vehicle-to-grid market bidding:** Reduced vehicle-availability forecast error by approximately 14% relative to the previous baseline, supporting capacity estimates for electricity-market bids. Selected PatchTST through model comparisons; tuned candidates with Python/PyTorch and Optuna on Vertex AI GPUs. Combined ERCOT prices via gridstatus, Monte Carlo capacity estimates, greedy vehicle selection, and load reallocation after disconnections.
- **Utility data onboarding (HOMeload):** Reduced manual electricity-data preparation time by approximately 50% with a Python/FastAPI pipeline combining Gemini schema discovery, pandas/lxml extraction, deterministic validation, and bounded corrective retries. Used BigQuery MERGE for repeatable loads; deployed with Docker, Cloud Run, and Terraform.
- **Rate recommendation:** Built utility-plan comparisons using XGBoost bill predictions from tariffs, vehicle telemetry, and household consumption. Delivered Flask APIs backed by BigQuery and a SAML-authenticated Dash interface on Cloud Run through Tekton CI/CD.
- **Energy data engineering and reporting:** Automated simulator-input preparation and optimizer-output processing with PySpark, BigQuery, and Airflow. Delivered Looker Studio discharge reporting and REST APIs for reviewing energy performance in operational dashboards.
- **Code analysis and cloud cost estimation:** Built a FastAPI/LangChain tool for engineering code reviews using GitPython, AST parsing, and Server-Sent Events. Generated LLM-assisted complexity estimates; calculated Vertex AI costs from assumed resources and static pricing.

CereVu Medical, Inc.

Software Developer / Research Engineer (Concurrent Contract) | Jun 2023 - Sep 2024

- **Wearable health platform:** Developed Vitality and Clinical, two SwiftUI apps for capturing and reviewing 9 physiological signals through CoreBluetooth BLE and Swift Charts. Migrated Clinical from UIKit to an MVVM-Coordinator architecture supporting calibration, exports, role-based workflows, and OAuth 2.0/JWT authentication.
- **Backend and delivery:** Connected wearable data to a React/Next.js dashboard through Python/Flask services, WebSocket streaming, and MongoDB. Deployed Docker services on AWS EC2 with GitHub Actions and distributed iOS builds through TestFlight.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Selected Projects

- **Business Action Prioritizer:** Designed a Google ADK/Gemini workflow that turns business data and scoring rubrics into ranked actions with supporting analysis. Gated PandasAI execution on user approval; implemented FastAPI/SQLite sessions and MCP report delivery through Cloud Storage.
- **InspecaIytics:** Reduced inspection-report preparation time by approximately 50% by connecting annotated photos directly to Word/PowerPoint reports using SwiftUI, React/TypeScript, and Supabase. Implemented offline edit conflict resolution, ARKit/Apple Vision photo organization, and XCTest/Vitest tests.
- **Clink Social:** Built SwiftUI/MapKit discovery and messaging with Firebase. Implemented TypeScript Cloud Functions for feeds and notifications, plus QR-based sign-in using time-limited OAuth/JWT tokens.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

Published research in machine learning and sensing; 500+ Google Scholar citations (September 2026).

- **Wearable stress detection:** Compared classifiers including XGBoost on windowed heart-rate/motion data from 54 students over 40 days, labeled by self-reports. Used Python, cross-validation, ROC-AUC evaluation, and SHAP interpretation.
- **MERCURY:** Co-authored research using random projection with quantization and cached computation reuse; reported 1.97x average training speedup across 12 models with accuracy similar to baseline.
- **OpenSync:** Co-developed open-source Python software using Lab Streaming Layer to synchronize sensors and experiment events; published tests measured RMS recording-time lag below 0.07 ms. GitHub repository.
- **Computer vision:** Developed face-mask and physical-distance monitoring using TensorFlow Faster R-CNN detectors and calibrated perspective transforms to measure distances in video.
- **GlucoseCoach:** Built an iOS application and Python/Flask backend for glucose monitoring.

Technical Skills

Core Languages: Python, SQL, TypeScript/JavaScript, Swift; additional: C++, Java, R, MATLAB

ML Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, OpenCV

ML Methods: Time-series forecasting, PatchTST, Monte Carlo simulation, cross-validation, SHAP/LIME

Generative AI: LangChain, Google ADK, Gemini, RAG, BM25, agent routing, text-to-SQL, MCP, PandasAI

Data: BigQuery, PySpark, Airflow, pandas, NumPy, PostgreSQL, MongoDB, Firebase, Supabase

Software Engineering: FastAPI, Flask, REST, WebSocket, React/Next.js, SwiftUI, CoreBluetooth, OAuth/JWT/SAML

Cloud: GCP (Vertex AI, Cloud Run, Dataproc), AWS (EC2, S3, Lambda, SageMaker), Azure OpenAI

Delivery and Testing: Docker, Terraform, Kubernetes, Git, GitHub Actions, Tekton, AWS CodeDeploy, Fastlane, TestFlight, XCTest, Vitest

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. JMIR Mental Health, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. IEEE HPCA, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. Journal of Neuroscience Methods, 2022.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. SN Computer Science, 2022.